

OMKAR KUDALKAR ↗

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EDUCATION

Dwarkadas Jivanlal Sanghvi College of Engineering

Sep 2024 – June 2028

B.Tech CSE (Data Science)

CGPA: 9.94

FIITJEE - Dixit Junior College of Science

2023 | 80%

N.L. Dalmia High School

2021 | 96.40%

TECHNICAL SKILLS

Languages: Python, C++, C, Java, JavaScript

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems

Backend & Databases: Node.js, Express.js, SQL, MongoDB, Neo4j, LangChain, LangGraph, REST APIs, Microservices

Frontend: Next.js, React.js, Three.js, Framer Motion, Tailwind CSS

Developer Tools & Practices: AWS, Docker, Kubernetes, Kafka, Git, GitHub, CI/CD, Unit Testing, Vercel, Netlify, Render, n8n

EXPERIENCE

Hooman Labs | *Founders Office Intern*

Apr 2026 – Present

- Built secure CRM integrations (HubSpot, Shopify, CleverTap) for enterprise clients (GIVA, Palmonas), linking APIs to build a unified outgoing-call analysis database and automate customer onboarding.
- Cut prompt token usage from 13–20K to under 5K per call by optimizing call-flow logic and prompt design, reducing per-call cost while maintaining conversation quality for clients like Careers360.
- Built QA metrics to analyze call recordings for disconnections and mid-call gaps, and collaborated with founders/engineers in an agile workflow to scope requirements and ship reliable fixes on tight timelines.

IIT Patna | *Research Intern* ↗

Apr 2026 – Present

- Architected an 11-agent parallel processing pipeline (Python, `asyncio.gather()`) with 3-wave orchestration, achieving 3–4× faster execution than sequential processing with fault-tolerant graceful degradation.
- Engineered a Graph-RAG retrieval pipeline using Neo4j subgraph extraction, cutting LLM context-window token usage by 86% (30K to under 6K tokens/query) through debugging and iterative tuning of retrieval logic.

ACHIEVEMENTS / AWARDS

- **1st Place (Winner) – HackXelerate 2026 (VJTI Mumbai)** **200+ Teams**
– Developed an AI-powered smart waste management platform using OpenCV for waste detection and intelligent routing to assign the nearest sanitation worker through a hybrid hardware-software architecture.
- **1st Place (Winner) – VCET 30-Hour Hackathon (Vidyavardhini College of Engineering)** **1500+ Teams**
– Built Code Guard, an AST + CFG static-analysis tool for path-sensitive resource-leak detection, as part of a 4-member team.
- **1st Place (Winner) – DevHacks 2026 (Atharva Engineering College)** **1000+ Participants**
– Built a Graph-RAG powered multi-agent storytelling platform featuring context-aware retrieval, AI comic generation, image visualization, and AI voice agents for automated story review and approval workflows.
- **Competitive Programming and DSA** **600+ Problems** – LeetCode, GeeksforGeeks, CodeChef
– 2★ rated on CodeChef; peak contest rating of **1509** on LeetCode.

PROJECTS

Code Guard ↗

Sep 2026

Python, AST/CFG Analysis, GitHub Actions, VS Code Extension, Docker, AWS

- Built a static-analysis engine using AST + control-flow graphs to trace every path from a resource's acquisition to release, flagging leaks like a DB connection opened but never closed on an exception path.
- Enforced leak-free code via a CI/CD gate: a pre-commit hook blocks the push and a Marketplace-listed GitHub Action fails the PR check when a leak is detected, alongside a VS Code extension for real-time coding-time detection and full-repo GitHub analysis with visual leak-path reports.
- Cut scan time by caching AST/CFG fingerprints in an evolving knowledge base to avoid redundant re-analysis of previously seen API-usage patterns; containerized the pipeline with Docker and deployed it on AWS for scalable, on-demand analysis.

AutoOps ↗

Sep 2026

Node.js, Express, Postgres, Redis, AWS SQS (LocalStack), Next.js, Docker

- Built an automated provisioning system that creates GitHub repos and Slack channels with correct permissions for approved requests, decoupling third-party API calls from the request path via an SQS-backed async queue.
- Cut API response latency to under 50ms (202 Accepted) by moving all GitHub/Slack provisioning calls off the request path onto the queue, isolating client-facing performance from third-party outages.
- Eliminated duplicate GitHub/Slack invites across worker crashes and message redeliveries with SHA-256 idempotency keys (Redis, 7-day TTL); automated recovery for failed jobs via a 3-attempt retry policy with dead-letter fallback and a frozen state machine with same-transaction audit logging, cutting manual intervention to a single click.

Nolan AI Studio ↗

March 2026

Neo4j, Next.js, LangChain, MongoDB, LangGraph

- Replaced chunking-based RAG with Graph-RAG using Neo4j to map character relationships and story entities as graph nodes, enabling faster retrieval and richer context-aware storytelling compared to vector similarity alone.
- Built a full professional UI with multi-agent workflows for AI comic generation, image visualization, and voice-agent narration, reducing average API calls per session by 3× through intelligent agent orchestration.